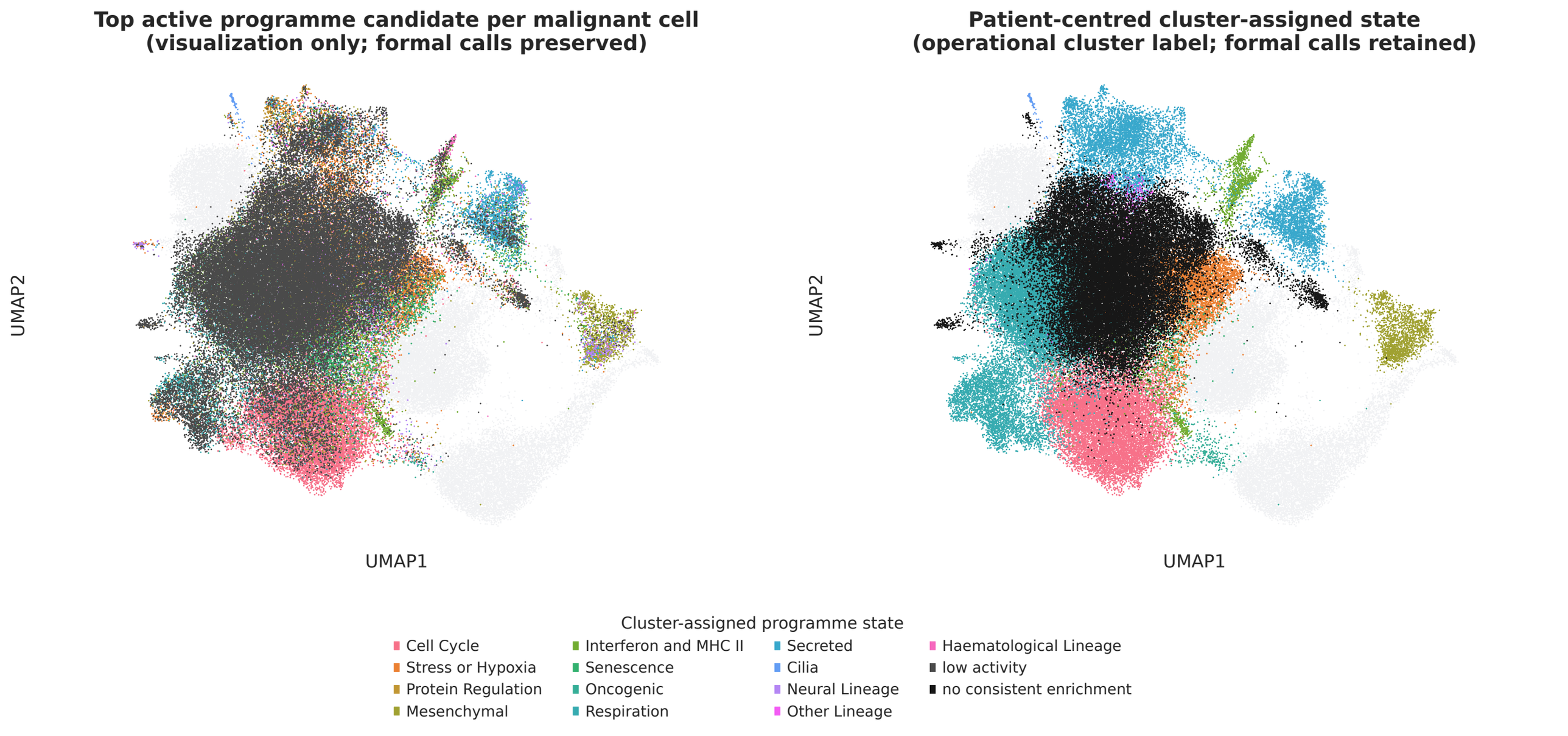

Top active programme candidate per malignant cell
(visualization only; formal calls preserved)
Patient-centred cluster-assigned state
(operational cluster label; formal calls retained)
UMAP2
UMAP1
UMAP2
UMAP1
Cluster-assigned programme state
Cell Cycle
Stress or Hypoxia
Protein Regulation
Mesenchymal
Interferon and MHC II
Senescence
Oncogenic
Respiration
Secreted
Cilia
Neural Lineage
Other Lineage
Haematological Lineage
low activity
no consistent enrichment